

Tirtho Mondal

📍 Kapilmuni, Khulna, Bangladesh ✉ tirthomondal.2001@gmail.com 📞 +8801571421684

🌐 LinkedIn 🐙 GitHub 🎓 Scholar 📄 ORCID 🌐 Website

EDUCATION

B.Sc. in Computer Science and Engineering

Khulna University of Engineering & Technology (KUET)

4th Year, 2nd Semester CGPA: 3.72 / 4.00

Higher Secondary Certificate (H.S.C.) in Science

Kapilmuni College

Board of Intermediate & Secondary Education, Jashore GPA: 5.00 / 5.00

RESEARCH INTERESTS

Machine Learning · Artificial Intelligence · Optimization Theory · Game Theory · Deep Learning · Computer Vision · Natural Language Processing

PUBLICATIONS

Real Time Assistive Navigation for the Visually Impaired: A Scalable Approach for Indoor and Outdoor Mobility

Dabbrata Das, Argho Deb Das, Farhan Sadaf, Azhar Uddin, **Tirtho Mondal**

arXiv preprint, arXiv:2504.20976 (2025) 🔗

RESEARCH & PROJECTS

Smart Navigation for the Visually Impaired 🗺️

- Real time assistive navigation combining computer vision and sensor fusion for safe, independent mobility.
- Implemented multimodal guidance using audio and haptic feedback for indoor and outdoor environments.

TutorFinder 🗺️

- Web platform connecting students with qualified tutors, featuring intelligent matching and scheduling.

HouseHold Services 🗺️

- Android application for booking and managing household services with real time status tracking.

CSE Family 🗺️

- Desktop platform for CSE students to share resources, discuss topics, and connect.

MobiCash 🗺️

- Mobile banking simulation supporting cash in, cash out, recharge, and balance management using OOP.

Laundry Shop Management 🗺️

- Relational database system designed to streamline laundry shop operations and customer tracking.

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, PHP, SQL, HTML/CSS, Java, Dart

Frameworks: Laravel, Android SDK, Flutter

Tools: Git/GitHub, VS Code, IntelliJ IDEA, Android Studio, Jupyter Notebook, LaTeX

Research Tools: NumPy, Pandas, OpenCV, Sensor Fusion

Concepts: Machine Learning, Deep Learning, Computer Vision, NLP, REST APIs, OOP, Design Patterns

Productivity: MS Word, Excel, PowerPoint

HONOURS & AWARDS

- **Dean's List:** Academic Years 2021–2022 and 2022–2023, Department of CSE, KUET

LANGUAGES

Bengali: Native **English:** Fluent (Academic & Professional)

PERSONAL STATEMENT

Self motivated final year CSE undergraduate with strong interest in research at the intersection of Machine Learning, Optimization, and Game Theory. Seeking research opportunities, graduate programs, or software engineering roles where I can apply and advance my skills in AI driven systems.